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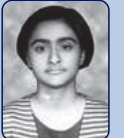
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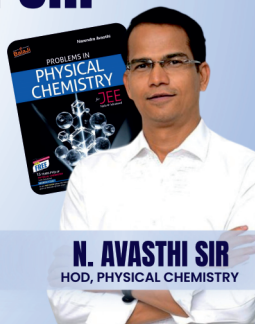
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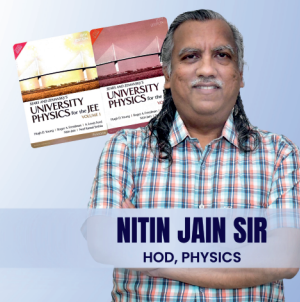


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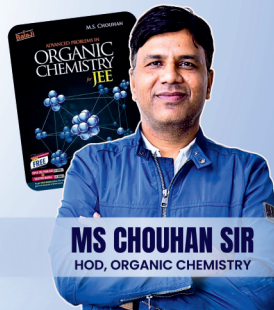


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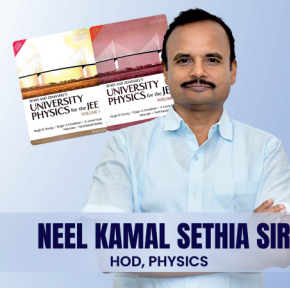


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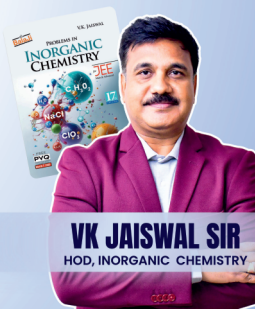


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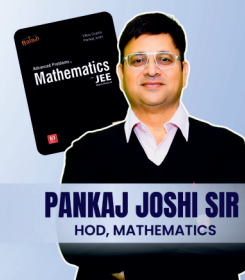


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PERIODIC PROPERTIES

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EXERCISE-I

History/Development Group/Period :

- Atomic weight of Cl = 35.5 and of I = 127. According to Doberiner triad rule, atomic weight of Br will be :
(1) 80.0 (2) 162.5 (3) 81.25 (4) 91.5
- There are 10 neutrons in the nucleus of the elements ${}_Z\text{M}^{19}$. It belongs to :
(1) f-block (2) s-block (3) d-block (4) none of these
- The elements of groups, 1, 2, 13, 14, 15, 16, 17 and 18 are collectively called -
(1) Noble gases (2) Representative or normal elements
(3) Transition elements (4) Inner transition elements
- In Lothar Meyer curve, the element in the peak of curve will be
(1) F (2) Na (3) Mg (4) Ne
- According to Mendeleev's periodic law, physical and chemical properties are function of
(1) Atomic number (2) Atomic weight (3) Atomic volume (4) Number of neutron
- Atomic number of Ag is 47. In the same group the atomic number of elements placed above and below Ag will be
(1) 37, 67 (2) 29, 79 (3) 39, 69 (4) 18, 28
- The number of elements in 5th and 6th period of periodic table are respectively -
(1) 8, 18 (2) 18, 18 (3) 18, 32 (4) 18, 28
- An element has atomic number 37. The block and group of this element are respectively :
(1) s and 1st (2) p and 17th (3) s and 2nd (4) p and 13th

Atomic size :

- Atomic radii of F & Ne in Angstrom are respectively given by :
(1) 0.72, 1.60 (2) 1.60, 1.60 (3) 0.72, 0.72 (4) 1.60, 0.72
- Which one of the following is the smallest in size ?
(1) N^{3-} (2) O^{2-} (3) F^- (4) Na^+
- Which of the following is the correct order of size of the given species
(1) $\text{I} > \text{I}^- > \text{I}^+$ (2) $\text{I}^+ > \text{I}^- > \text{I}$ (3) $\text{I} > \text{I}^+ > \text{I}^-$ (4) $\text{I}^- > \text{I} > \text{I}^+$
- The correct order of size would be -
(1) $\text{Ni} < \text{Pd} \approx \text{Pt}$ (2) $\text{Pd} < \text{Pt} < \text{Ni}$ (3) $\text{Pt} > \text{Ni} > \text{Pd}$ (4) $\text{Pd} > \text{Pt} > \text{Ni}$
- Arrange the following in order of increasing atomic radii Na, Si, Al, Ar -
(1) $\text{Na} < \text{Si} < \text{Al} < \text{Ar}$ (2) $\text{Si} < \text{Al} < \text{Na} < \text{Ar}$
(3) $\text{Ar} < \text{Al} < \text{Si} < \text{Na}$ (4) $\text{Na} < \text{Al} < \text{Si} < \text{Ar}$
- Arrange the elements in increasing order of atomic radius Na, Rb, K, Mg -
(1) $\text{Na} < \text{K} < \text{Mg} < \text{Rb}$ (2) $\text{K} < \text{Na} < \text{Mg} < \text{Rb}$
(3) $\text{Mg} < \text{Na} < \text{K} < \text{Rb}$ (4) $\text{Rb} < \text{K} < \text{Mg} < \text{Na}$

15. Following triads have approximately equal size

- (1) Na^+ , Mg^{2+} , Al^{3+} (isoelectronic) (2) F^- , Ne , O^{2-} (isoelectronic)
(3) Fe, Co, Ni (4) Mn^+ , Fe^{2+} , Cr (iso-electronic)

Ionisation Energy :

16. Correct orders of 1st I.P. are -

- (i) $\text{Li} < \text{B} < \text{Be} < \text{C}$ (ii) $\text{O} < \text{N} < \text{F}$ (iii) $\text{Be} < \text{N} < \text{Ne}$
(1) (i), (ii) (2) (ii), (iii) (3) (i), (iii) (4) (i), (ii), (iii)

17. Ionisation energy increases in the order -

- (1) $\text{Be} < \text{B} < \text{C} < \text{N}$ (2) $\text{B} < \text{Be} < \text{C} < \text{N}$ (3) $\text{C} < \text{N} < \text{Be} < \text{B}$ (4) $\text{N} < \text{C} < \text{Be} < \text{B}$

18. Highest ionisation potential in a period is shown by -

- (1) Alkali metals (2) Noble gases (3) Halogens (4) Representative elements

19. The first ionisation energy will be maximum for :

- (1) Be (2) He (3) Li (4) Fe

20. Ionisation energy :

- (1) increases with an increase in atomic radii.
(2) is independent of atomic radii.
(3) decreases with an increase in atomic radii.
(4) remains constant with an increase or decrease in atomic radii.

21. The first ionisation energy of Na, Mg, Al and Si are in the order :

- (1) $\text{Na} < \text{Mg} < \text{Al} < \text{Si}$ (2) $\text{Na} < \text{Al} < \text{Mg} < \text{Si}$
(3) $\text{Mg} < \text{Na} < \text{Al} < \text{Si}$ (4) $\text{Si} < \text{Mg} < \text{Al} < \text{Na}$

22. Which of the following is the correct order of ionization energy ?

- (1) $\text{O}^{2-} < \text{F}^- < \text{Na}^+ < \text{Mg}^{2+}$ (2) $\text{F}^- < \text{O}^{2-} < \text{Na}^+ < \text{Mg}^{2+}$
(3) $\text{O}^{2-} < \text{Na}^+ < \text{F}^- < \text{Mg}^{2+}$ (4) $\text{Mg}^{2+} < \text{Na}^+ < \text{F}^- < \text{O}^{2-}$

23. The value of IE(I), IE(II), IE(III), and IE(IV), of an atom are respectively 7.5 eV, 25.6 eV, 48.6 eV and 170.6 eV. The electronic configuration of the atom will be :

- (1) $1s^2, 2s^2, 2p^6, 3s^1$ (2) $1s^2, 2s^2, 2p^6, 3s^2, 3p^1$
(3) $1s^2, 2s^2, 2p^6, 3s^2, 3p^3$ (4) $1s^2, 2s^2, 2p^6, 3s^2$

24. IP_1 and IP_2 of Mg are 178 and 348 K.cal mol⁻¹. The enthalpy required for the reaction $\text{Mg} \rightarrow \text{Mg}^{2+} + 2e^-$ is -

- (1) + 170 K.cal (2) + 526 K.cal (3) - 170 K.cal (4) - 526 K.cal

Electron Gain Enthalpy, Electron affinity :

25. In the process $\text{Cl}_{(g)} + e^- \xrightarrow{\Delta H} \text{Cl}^-(g)$, ΔH is -

- (1) Positive (2) Negative (3) Zero (4) None

26. Which of the following process energy is liberated -

- (1) $\text{Cl} \rightarrow \text{Cl}^+ + e^-$ (2) $\text{HCl} \rightarrow \text{H}^+ + \text{Cl}^-$ (3) $\text{Cl} + e^- \rightarrow \text{Cl}^-$ (4) $\text{O}^- + e^- \rightarrow \text{O}^{2-}$

27. Second electron affinity of an element is -

- (1) Always exothermic (2) Endothermic for few elements
(3) Exothermic for few elements (4) Always endothermic

28. Which of the following configuration will have least electron affinity -
 (1) ns^2np^5 (2) ns^2np^2 (3) ns^2np^3 (4) ns^2np^4
29. Which of the following will have the most negative electron gain enthalpy and which the least negative?
 (1) F, Cl (2) Cl, F (3) S, Cl (4) Cl, P
30. The electron affinity of the members of oxygen of the periodic table, follows the sequence
 (1) $O > S > Se$ (2) $S > O > Se$ (3) $O < S > Se$ (4) $Se > O > S$
31. Among halogens, the correct order of amount of energy released in electron gain (electron gain enthalpy) is
 (1) $F > Cl > Br > I$ (2) $F < Cl < Br < I$ (3) $F < Cl > Br > I$ (4) $F < Cl < Br < I$
32. The electronic affinity values (in kJ mol^{-1}) of three halogens X, Y and Z are respectively – 349, – 333 and – 325. Then X, Y and Z respectively are
 (1) F, Cl and Br (2) Cl, F and Br (3) Cl, Br and F (4) Br, Cl and F

Electronegativity :

33. The electronegativity values of C, N, O and F :
 (1) increase from carbon to fluorine. (2) decrease from carbon to fluorine.
 (3) increase up to oxygen and is minimum at fluorine.
 (4) is minimum at nitrogen and the increase continuously.
34. Electronegativity of an element is 1.0 on the Pauling scale. Its value on Mulliken scale
 (1) 2.8 (2) 1 (3) 2.0 (4) 1.5
35. In the following which configuration of element has maximum electronegativity.
 (1) $1s^2, 2s^2 2p^5$ (2) $1s^2, 2s^2 2p^6$ (3) $1s^2, 2s^2 2p^4$ (4) $1s^2, 2s^2 2p^6, 3s^2 3p^3$
36. Which of the following is affected by the stable electron configuration of an atom ?
 (a) electronegativity (b) Ionisation energy (c) Electron affinity
 Correct answer is :
 (1) Only electronegativity (2) only ionisation potential
 (3) electron affinity and ionisation energy both (4) All of the above
37. Following the Mulliken scale, what parameters are required to evaluate electronegativity ?
 (1) Only electronegativity (2) Only electron affinity
 (3) Electron affinity and ionization energy (4) Ionic potential and electronegativity
38. Electronegativity of the following elements increases in the order -
 (1) $O < N < S < P$ (2) $P < S < N < O$
 (3) $P < N < S < O$ (4) $S < P < N < O$

Miscellaneous :

39. Which of the following elements do not belong to the family indicated ?
 (1) $_{29}\text{Cu}$ - Coinage metal (2) $_{56}\text{Ba}$ - Alkaline earth metal
 (3) $_{30}\text{Zn}$ - Alkaline earth metal (4) $_{54}\text{Xe}$ - Noble gas
40. Zr and Hf have nearly same size because
 (1) They belong to same group (2) They belong to same period
 (3) Of lanthanoid contraction (4) Of poor screening of d orbital

41. The elements in which electrons are progressively filled in 4f-orbital are called
 (1) actinoids (2) transition elements (3) lanthanoids (4) halogens
42. Which of the following relation is correct with respect to first (I) and second (II) ionisation energies of sodium and magnesium ?
 (1) $I_{\text{mg}} = II_{\text{Na}}$ (2) $I_{\text{Na}} > I_{\text{Mg}}$ (3) $II_{\text{Mg}} > II_{\text{Na}}$ (4) $II_{\text{Na}} > II_{\text{Mg}}$
43. Correct order of Ionisation energy is
 (1) $B > Tl > Ga > Al > In$ (2) $B > Al > Ga > In > Tl$
 (3) $He > Ar > Ne > Kr > Xe$ (4) $Li > K > Na > Rb > Cs$
44. Amphoteric behaviour is shown by the oxides of :
 (1) Al and Ca (2) Pb and Ba (3) Cr and Mg (4) Sn and Zn
45. The oxide which have highest acidic character
 (1) MnO (2) MnO_2 (3) Mn_2O_3 (4) Equal in all of these
46. An element X occurs in short period having configuration ns^2np^1 . The formula and nature of its oxide is :
 (1) XO_3 , basic (2) XO_3 , acidic (3) X_2O_3 , amphoteric (4) X_2O_3 , basic
47. The element having very high ionization enethalpy but zero electron affinity is-
 (1) H (2) F (3) He (4) Be
48. Correct order of electronegativity of N, P, C and Si is -
 (1) $N < P < C < Si$ (2) $N > C > Si > P$
 (3) $N = P > C = Si$ (4) $N > C > P > Si$
49. The process requiring the absorption of energy is -
 (1) $F \rightarrow F^-$ (2) $Cl \rightarrow Cl^-$ (3) $O \rightarrow O^{2-}$ (4) $H \rightarrow H^-$
50. Atomic number 15, 33, 51 represents the following family -
 (1) Carbon family (2) Nitrogen family (3) Oxygen family (4) None
51. The name 'Rare earths' is used for -
 (1) Lanthanides only (2) Actinides only
 (3) Both lanthanides and actinides (4) Alakaline earth metals
52. Which are correct match -
 (i) Eka silicon – Be (ii) Eka aluminium – Ga
 (iii) Eka manganese – Tc (iv) Eka scandium – B
 (1) (ii) & (iii) (2) (i), (ii) & (iv) (3) (i) & (iv) (4) All
53. The places that were left empty by Mendeleef were, for -
 (1) Aluminium & Silicon (2) Gallium and germinium
 (3) Arsenic and antimony (4) Molybdenum and tungsten
54. If x, y and z are electronegativity, ionisation potential and electron-affinity respectively. Then the electron affinity (z) in the terms of electronegativity (x) and ionisation potential (y) will be :
 (1) $z = \frac{x+y}{2}$ (2) $z = \frac{x-y}{2}$ (3) $z = \frac{x^2 - y^2}{2}$ (4) $z = 2x - y$

55. The element which have highest 2nd ionisation energy
 (1) Na (2) Mg (3) Ca (4) Ar
56. When the following five anions are arranged in order of decreasing ionic radius, the correct sequence is :
 (1) Se^{2-} , I^- , Br^- , O^{2-} , F^- (2) I^- , Se^{2-} , O^{2-} , Br^- , F^- (3) Se^{2-} , I^- , Br^- , F^- , O^{2-} (4) I^- , Se^{2-} , Br^- , O^{2-} , F^-
57. The alkali metal which is radioactive
 (1) Fr (2) Al (3) Li (4) Mg
58. The order of screening effect of electrons of s, p, d and f orbitals of a given shell of an atom on its outer shell electrons is
 (1) $s > p > d > f$ (2) $f > d > p > s$ (3) $p < d < s > f$ (4) $f > p > s > d$

Integer types Questions :

59. Find out total number of representative elements in the given elements :
 Cd, Nb, Ta, Te, Ra, Mo, Po, Tc
60. How many pairs are, in which first species has lower ionisation energy than second species :
 (i) N and O (ii) Br and K (iii) Be and B (iv) I and I^-
 (v) Li and Li^+ (vi) O and S (vii) Ba and Sr
61. Calculate the value of $\frac{x+y}{2}$
 where;
 x = Number of 'd' electrons retained in Fe^{2+} ion
 y = Number of 'd' electrons retained in Cr^+ ion
62. Calculate the value of $\frac{x+y}{2}$
 where x = number of elements having higher I.E. than Cl
 y = number of elements having lower I.E. than Cl
 Given elements are :
 K, Na, Ne, Mg, Al, Si, P, S, Rb, Ar, He
63. Find the sum of most stable oxidation state of Eka silicon and Eka Aluminium which were predicted by Mendeleev now present in p-block in long form of periodic table.
64. Find out total number of anions which are bigger in size than S^{2-} ion.
 P^{3-} , As^{3-} , O^{2-} , Se^{2-} , Te^{2-} , F^- , I^- , Cl^-
65. The number of electrons for Zn^{2+} cation that have value of Azimuthal quantum number, $\ell = 0$ is :
66. Select total number of pair(s) in which Ionisation energy of first element is less than Ionisation energy of second element
 (Na, K), (Li, Be), (He, Ne), (N, O), (In, Tl), (Zn, Cd),
 (Ag, Au), (Al, Ga), (Sn, Pb)
67. In how many pairs the second species have larger size than the first one ?
 (F, Cl), (Na, Na^+), (Cu, Zn), (Mg, Al), (Cl^- , Cl)
68. If the first, second and third I.E. values of an element are 100 eV, 150 eV and 1500 eV. Then the number of valence electron in the element is

EXERCISE-II

- The element with atomic number 57 belongs to :
(1) s-block (2) p-block (3) d-block (4) f-block
- Element in periodic table with electronic configuration as $[\text{Ar}] 3d^5 4s^1$ is placed in :
(1) 1st group, s-block (2) 2nd group, s-block
(3) 5th group, d-block (4) 6th group, d-block
- Choose the incorrect option for the order of atomic radius :
(1) $\text{Au} \approx \text{Ag} < \text{Cu}$ (2) $\text{Li} < \text{Ne}, \text{Na} < \text{Ar}$
(3) $\text{O}^{2-} > \text{O}^- > \text{O} > \text{O}^+$ (4) $\text{Sc} < \text{Y} < \text{La}$
- Values of IE_1 , IE_2 and IE_3 of an element are 9.32, 18.21 and 553.83 eV. What informations do these data convey ?
(1) The element has two electrons in the valence shell.
(2) The element has two p-electrons in the valence shell.
(3) (1) and (2) both.
(4) None of the two.
- Which of the following order is **correct**
(1) $\text{S}^{2-} < \text{Cl}^- < \text{K}^+$: order of size (2) $\text{K} < \text{Na} < \text{Mg} < \text{Ne}$: order of ionisation energy
(3) $\text{Mg} < \text{Al} < \text{Si}$: order ionisation energy (4) $\text{Mg}^{2+} < \text{Al}^{3+} < \text{Na}^+$: order of size
- The correct order of increasing atomic radius of the following elements is :
(1) $\text{S} < \text{O} < \text{Se} < \text{C}$ (2) $\text{O} < \text{C} < \text{S} < \text{Se}$ (3) $\text{O} < \text{S} < \text{Se} < \text{C}$ (4) $\text{C} < \text{O} < \text{S} < \text{Se}$
- Amongst the following elements, the highest ionisation energy is :
(1) $[\text{Ne}] 3s^2 3p^1$ (2) $[\text{Ne}] 3s^2 3p^4$ (3) $[\text{Ne}] 3s^2 3p^3$ (4) $[\text{Ne}] 3s^2 3p^5$
- Which of the following statements are correct ?
(a) HF is a stronger acid than HCl.
(b) among halide ions, iodide is the most powerful reducing agent.
(c) Fluorine is the only halogen that does not show a variable oxidation state.
(d) HOCl is a stronger acid than HOBr.
(1) b and d (2) b and c (3) a, b and c (4) b, c, and d
- If the same element is forming oxides in different oxidation states then :
(1) that oxide will be neutral in nature in which element will be in its highest oxidation state.
(2) that oxide will be highest acidic in nature in which element will be in its highest oxidation state.
(3) that oxide will be amphoteric in nature in which element will be in its highest oxidation state.
(4) that oxide will be highly basic in nature in which element will be in its highest oxidation state.
- Elements with atomic number 58 to 71 are placed in :
(1) 5th period and 4th group (2) 6th period and 3th group
(3) 7th period and 3th group (4) 6th period and 4th group

11. Which of the following is not concerned to effective nuclear charge?
 - (1) Higher ionization potential of carbon than boron
 - (2) Higher ionization potential of magnesium than aluminium
 - (3) Higher values of successive ionization energy
 - (4) Higher electronegativity of Mn^{7+} than Mn^{4+}
12. Which one of the following oxide have highest acidic character ?
 - (1) CO_2
 - (2) ClO_2
 - (3) SiO_2
 - (4) SO_2
13. Which is a true statement ?
 - (1) Larger is the value of ionisation energy easier is the formation of cation.
 - (2) Larger is the value of electron affinity easier is the formation of anion.
 - (3) Larger is the value of ionisation energy as well as electron affinity the smaller is the electronegativity of atom.
 - (4) Larger is the Z_{eff} larger is the size of atom.
14. Electron gain enthalpy will be positive in
 - (1) O^{2-} is formed from O^{-1}
 - (2) O^{1-} is formed from O
 - (3) S^{-1} is formed from S
 - (4) Na^{-} is formed from Na
15. The element which has highest electron affinity ?
 - (1) Oxygen
 - (2) Sulphur
 - (3) Nitrogen
 - (4) Phosphorus
16. An atom with high electronegativity generally has a
 - (1) Low electron affinity
 - (2) Small atomic number
 - (3) Large atomic radius
 - (4) High ionisation potential
17. For electron affinity of halogens which of the following is correct ?
 - (1) $Br > F$
 - (2) $F > Cl$
 - (3) $Br < Cl$
 - (4) $F < I$
18. The correct order of negative electron gain enthalpy is :
 - (1) $Be < B < C < N$
 - (2) $Be < N < B < C$
 - (3) $N < Be < C < B$
 - (4) $N < C < B < Be$
19. Increasing order of second ionization energy is
 - (1) $Ne < O < F < N < B < C < Be$
 - (2) $Be < C < B < N < F < O < Ne$
 - (3) $Be < B < C < N < F < O < Ne$
 - (4) $B < C < N < O < F < Be < Ne$
20. The first ionisation potentials of Na, Mg, Al and Si are in the order
 - (1) $Na < Mg > Al < Si$
 - (2) $Na < Mg < Al > Si$
 - (3) $Na > Mg > Al > Si$
 - (4) $Na > Mg > Al < Si$
21. Which of the following is wrong -
 - (1) $NH_3 < PH_3 < AsH_3 \Rightarrow$ Acidic character
 - (2) $Li < Be < B < C \Rightarrow IE_1$
 - (3) $Al_2O_3 < MgO < Na_2O < K_2O \Rightarrow$ Basic character
 - (4) $Li^+ < Na^+ < K^+ < Cs^+ \Rightarrow$ Ionic radius
22. Which of the following is correct order of increasing bond strength -
 - (1) $HF < HCl < HBr < HI$
 - (2) $HI < HBr < HF < HCl$
 - (3) $HI < HBr < HCl < HF$
 - (4) $HCl < HBr < HF < HI$
23. Density of sodium and potassium follows the order -
 - (1) $K < Na$
 - (2) $K > Na$
 - (3) $Na \approx K$
 - (4) None of these
24. The size of the species, Pb, Pb^{2+} , Pb^{4+} decreases as -
 - (1) $Pb^{4+} > Pb^{2+} > Pb$
 - (2) $Pb > Pb^{2+} > Pb^{4+}$
 - (3) $Pb > Pb^{4+} > Pb^{2+}$
 - (4) $Pb^{4+} > Pb > Pb^{2+}$

EXERCISE-III

(Question asked in previous JEE-MAIN)

1. B has a smaller first ionization enthalpy than Be. Consider the following statements : [JEE-Main-Online 2020]
- (I) It is easier to remove 2p electron than 2s electron.
(II) 2p electron of B is more shielded from the nucleus by the inner core of electrons than the 2s electrons of Be.
(III) 2s electrons has more penetration power than 2p electron.
(IV) Atomic radius of B is more than Be (atomic number B = 5, Be = 4)
The correct statements are :
- (1) (II), (III) and (IV) (2) (I), (III) and (IV) (3) (I), (II) and (IV) (4) (I), (II) and (III)
2. The acidic, basic and amphoteric oxides, respectively, are : [JEE-Main-Online 2020]
- (1) MgO, Cl₂O, Al₂O₃ (2) Na₂O, SO₃, Al₂O₃ (3) N₂O₃, Li₂O, Al₂O₃ (4) Cl₂O, CaO, P₄O₁₀
3. The first ionization energy (in kJ/mol) of Na, Mg, Al and Si respectively, are : [JEE-Main-Online 2020]
- (1) 496, 577, 786, 737 (2) 786, 737, 577, 496
(3) 496, 737, 577, 786 (4) 496, 577, 737, 786
4. The increasing order of the atomic radii of the following elements is: [JEE-Main-Online 2020]
- (a) C (b) O (c) F (d) Cl (e) Br
- (1) (a) < (b) < (c) < (d) < (e) (2) (d) < (c) < (b) < (a) < (e)
(3) (c) < (b) < (a) < (d) < (e) (4) (b) < (c) < (d) < (a) < (e)
5. The atomic radius of Ag is closest to : [JEE-Main-Online 2020]
- (1) Hg (2) Au (3) Cu (4) Ni
6. Within each pair of elements F & Cl, S & Se, and Li & Na, respectively, the elements that release more energy upon an electron gain are : [JEE-Main-Online 2020]
- (1) Cl, S and Li (2) F, Se and Na (3) Cl, Se and Na (4) F, S and Li
7. The electron gain enthalpy (in kJ/mol) of fluorine, chlorine, bromine and iodine, respectively, are : [JEE-Main-Online 2020]
- (1) -333, -325, -349 and -296 (2) -333, -349, -325 and -296
(3) -296, -325, -333 and -349 (4) -349, -333, -325 and -296
8. In general, the property (magnitudes only) that shows an opposite trend in comparison to other properties across a period is : [JEE-Main-Online 2020]
- (1) Ionization enthalpy (2) Atomic radius
(3) Electronegativity (4) Electron gain enthalpy
9. Three elements X, Y and Z are in the 3rd period of the periodic table. The oxides of X, Y and Z, respectively, are basic, amphoteric and acidic. The correct order of the atomic numbers of X, Y and Z is : [JEE-Main-Online 2020]
- (1) Z < Y < X (2) X < Y < Z (3) X < Z < Y (4) Y < X < Z
10. The number of subshells associated with n = 4 and m = -2 quantum numbers is : [JEE-Main-Online 2020]
- (1) 8 (2) 2 (3) 16 (4) 4

11. The atomic number of the element unnilennium is : **[JEE-Main-Online 2020]**
 (1) 109 (2) 102 (3) 108 (4) 119
12. Among the statements (I – IV), the correct ones are: **[JEE-Main-Online 2020]**
 (I) Be has smaller atomic radius compared to Mg.
 (II) Be has higher ionization enthalpy than Al.
 (III) Charge/radius ratio of Be is greater than that of Al.
 (IV) Both Be and Al form mainly covalent compounds.
 (1) (I), (II) and (IV) (2) (I), (III) and (IV) (3) (II), (III) and (IV) (4) (I), (II) and (III)
13. Consider the hypothetical situation where the azimuthal quantum number, ℓ , takes values 0, 1, 2, $n + 1$, where n is the principal quantum number. Then, the element with atomic number : **[JEE-Main-Online 2020]**
 (1) 9 is the first alkali metal (2) 13 has a half-filled valence subshell
 (3) 8 is the first noble gas (4) 6 has a 2p-valence subshell
14. The five successive ionization enthalpies of an element are 800, 2427, 3658, 25024 and 32824 kJ mol⁻¹. The number of valence electrons in the element is: **[JEE-Main-Online 2020]**
 (1) 5 (2) 4 (3) 3 (4) 2
15. The element with atomic numbers 101 and 104 belong to, respectively **[JEE-Main-Online 2020]**
 (1) Group 11 and Group 4 (2) Actinoids and Group 6
 (3) Actinoids and Group 4 (4) Group 6 and Actinoids
16. The ionic radii of O²⁻, F⁻, Na⁺ and Mg²⁺ are in the order : **[JEE-Main-Online 2020]**
 (1) F⁻ > O²⁻ > Na⁺ > Mg²⁺ (2) O²⁻ > F⁻ > Na⁺ > Mg²⁺
 (3) Mg²⁺ > Na⁺ > F⁻ > O²⁻ (4) O²⁻ > F⁻ > Mg²⁺ > Na⁺
17. The process that is NOT endothermic in nature is: **[JEE-Main-Online 2020]**
 (1) $\text{Ar}_{(g)} + e^- \longrightarrow \text{Ar}_{(g)}^-$ (2) $\text{H}_{(g)} + e^- \longrightarrow \text{H}_{(g)}^-$ (3) $\text{O}_{(g)}^- + e^- \longrightarrow \text{O}_{(g)}^{2-}$ (4) $\text{Na}_{(g)} \longrightarrow \text{Na}_{(g)}^+ + e^-$
18. In the sixth period, the orbitals that are filled are : **[JEE-Main-Online 2020]**
 (1) 6s, 4f, 5d, 6p (2) 6s, 5d, 5f, 6p (3) 6s, 5p, 6d, 6p (4) 6s, 6p, 6d, 6f
19. The correct electronic configuration and spin-only magnetic moment (BM) of Gd³⁺ (Z = 64), respectively, are : **[JEE-Main-Online 2020]**
 (1) [Xe] 4f⁷ and 8.9 (2) [Xe] 4f⁷ and 7.9 (3) [Xe] 5f⁷ and 8.9 (4) [Xe] 5f⁷ and 7.9
20. The correct order of the ionic radii of O²⁻, N³⁻, F⁻, Mg²⁺, Na⁺ and Al³⁺ is : **[JEE-Main-Online 2020]**
 (1) N³⁻ < O²⁻ < F⁻ < Na⁺ < Mg²⁺ < Al³⁺ (2) Al³⁺ < Na⁺ < Mg²⁺ < O²⁻ < F⁻ < N³⁻
 (3) Al³⁺ < Mg²⁺ < Na⁺ < F⁻ < O²⁻ < N³⁻ (4) N³⁻ < F⁻ < O²⁻ < Mg²⁺ < Na⁺ < Al³⁺
21. Lattice enthalpy and enthalpy of solution of NaCl are 788 kJ mol⁻¹ and 4 kJ mol⁻¹, respectively. The hydration enthalpy of NaCl is : **[JEE-Main-Online 2020]**
 (1) -780 kJ mol⁻¹ (2) 780 kJ mol⁻¹ (3) -784 kJ mol⁻¹ (4) 784 kJ mol⁻¹
22. The set that contains atomic numbers of only transition elements, is : **[JEE-Main-Online 2020]**
 (1) 37, 42, 50, 64 (2) 21, 25, 42, 72 (3) 9, 17, 34, 38 (4) 21, 32, 53, 64

23. The atomic number of Unnilunium is? [JEE-Main-Online 2020]
24. In which of the following pairs, the outer most electronic configuration will be the same? [JEE-Main-Online 2021]
 (1) Cr^+ and Mn^{2+} (2) Fe^{2+} and Co^+ (3) Ni^{2+} and Cu^+ (4) V^{2+} and Cr^+
25. Match List-I with List-II

List-I	List-II
Electronic configuration of elements	$\Delta_f H$ in kJ mol^{-1}
(a) $1s^2 2s^2$	(i) 801
(b) $1s^2 2s^2 2p^4$	(ii) 899
(c) $1s^2 2s^2 2p^3$	(iii) 1314
(d) $1s^2 2s^2 2p^1$	(iv) 1402

 Choose the most appropriate answer [JEE-Main-Online 2021]
 (1) (a) $\rightarrow$ (i), (b) $\rightarrow$ (iv), (c) $\rightarrow$ (iii), (d) $\rightarrow$ (ii)
 (2) (a) $\rightarrow$ (i), (b) $\rightarrow$ (iii), (c) $\rightarrow$ (iv), (d) $\rightarrow$ (ii)
 (3) (a) $\rightarrow$ (ii), (b) $\rightarrow$ (iii), (c) $\rightarrow$ (iv), (d) $\rightarrow$ (i)
 (4) (a) $\rightarrow$ (iv), (b) $\rightarrow$ (i), (c) $\rightarrow$ (ii), (d) $\rightarrow$ (iii)
26. Which pair of oxides is acidic in nature? [JEE-Main-Online 2021]
 (1) CaO , SiO_2 (2) B_2O_3 , CaO (3) N_2O , BaO (4) B_2O_3 , SiO_2
27. The correct order of electron gain enthalpy is [JEE-Main-Online 2021]
 (1) $\text{S} > \text{Se} > \text{Te} > \text{O}$ (2) $\text{Te} > \text{Se} > \text{S} > \text{O}$ (3) $\text{S} > \text{O} > \text{Se} > \text{Te}$ (4) $\text{O} > \text{S} > \text{Se} > \text{Te}$
28. Consider the elements Mg, Al, S, P and Si, the correct increasing order of their first ionization enthalpy is [JEE-Main-Online 2021]
 (1) $\text{Mg} < \text{Al} < \text{Si} < \text{S} < \text{P}$ (2) $\text{Al} < \text{Mg} < \text{Si} < \text{S} < \text{P}$
 (3) $\text{Mg} < \text{Al} < \text{Si} < \text{P} < \text{S}$ (4) $\text{Al} < \text{Mg} < \text{S} < \text{Si} < \text{P}$
29. Number of amphoteric compounds among the following is _____. [JEE-Main-Online 2021]
 (A) BeO (B) BaO (3) $\text{Be}(\text{OH})_2$ (D) $\text{Sr}(\text{OH})_2$
30. The spin only magnetic moment of a divalent ion in aqueous solution (atomic number 29) is _____ BM. [JEE-Main-Online 2021]
31. The ionization enthalpy of Na^+ formation from $\text{Na}_{(g)}$ is $495.8 \text{ kJ mol}^{-1}$, while the electron gain enthalpy of Br is $-325.0 \text{ kJ mol}^{-1}$. Given the lattice enthalpy of NaBr is $-728.4 \text{ kJ mol}^{-1}$. The energy for the formation of NaBr ionic solid is $(-) \text{ _____} \times 10^{-1} \text{ kJ mol}^{-1}$. [JEE-Main-Online 2021]
32. Identify the elements X and Y using the ionisation energy values given below :
 Ionization energy (kJ/mol)
- | | 1st | 2nd |
|---|-----|------|
| X | 495 | 4563 |
| Y | 731 | 1450 |
- (1) $\text{X} = \text{Na}$; $\text{Y} = \text{Mg}$ (2) $\text{X} = \text{Mg}$; $\text{Y} = \text{F}$ (3) $\text{X} = \text{Mg}$; $\text{Y} = \text{Na}$ (4) $\text{X} = \text{F}$; $\text{Y} = \text{Mg}$ [JEE-Main-Online 2021]

33. The characteristics of elements X, Y and Z with atomic numbers, respectively, 33, 53 and 83 are :
[JEE-Main-Online 2021]
(1) X and Y are metalloids and Z is a metal.
(2) X is a metalloid, Y is a non metal Z is a metal
(3) X, Y and Z are metals.
(4) X and Z are non metals and Y is a metalloid.
34. Arrange the following metal complex / compounds in the increasing order of spin only magnetic moment. Presume all the three, high spin system.
[JEE-Main-Online 2021]
(Atomic numbers Ce = 58, Gd = 64 and Eu = 63)
(a) $(\text{NH}_4)_2[\text{Ce}(\text{NO}_3)_6]$ (b) $\text{Gd}(\text{NO}_3)_3$ and (c) $\text{Eu}(\text{NO}_3)_3$
(1) (b) < (a) < (c) (2) (c) < (a) < (b) (3) (a) < (b) < (c) (4) (a) < (c) < (b)
35. The absolute value of the electron gain enthalpy of halogens satisfies : [JEE-Main-Online 2021]
(1) $\text{I} > \text{Br} > \text{Cl} > \text{F}$ (2) $\text{Cl} > \text{Br} > \text{F} > \text{I}$ (3) $\text{Cl} > \text{F} > \text{Br} > \text{I}$ (4) $\text{F} > \text{Cl} > \text{Br} > \text{I}$
36. What is the spin-only magnetic moment value (BM) of a divalent metal ion with atomic number 25, in its aqueous solution ? [JEE-Main-Online 2021]
(1) 5.92 (2) 5.0 (3) Zero (4) 5.26
37. The oxide that shows magnetic property is : [JEE-Main-Online 2021]
(1) SiO_2 (2) Mn_3O_4 (3) Na_2O (4) MgO
38. The first ionization energy of magnesium is smaller as compared to that of elements X and Y, but higher than that of Z. The elements X, Y and Z, respectively, are : [JEE-Main-Online 2021]
(1) chlorine, lithium and sodium (2) argon, lithium and sodium
(3) argon, chlorine and sodium (4) neon, sodium and chlorine
39. The set of elements that differ in mutual relationship from those of the other sets is : [JEE-Main-Online 2021]
(1) Li-Mg (2) B – Si (3) Be – Al (4) Li – Na
40. The ionic radius of Na^+ ions is 1.02 Å. The ionic radii (in Å) of Mg^{2+} and Al^{3+} , respectively, are : [JEE-Main-Online 2021]
(1) 1.05 and 0.99 (2) 0.72 and 0.54 (3) 0.85 and 0.99 (4) 0.68 and 0.72
41. In the ground state of atomic Fe ($Z = 26$), the spin-only magnetic moment is _____ $\times 10^{-1}$ BM. (Round off to the Nearest Integer).
[Given : $\sqrt{3} = 1.73$, $\sqrt{2} = 1.41$] [JEE-Main-Online 2021]
42. Outermost electronic configuration of a group 13 element, E, is $4s^2, 4p^1$. The electronic configuration of an element of p-block period-five placed diagonally to element, E is : [JEE-Main-Online 2021]
(1) $[\text{Xe}] 5d^{10} 6s^2 6p^2$ (2) $[\text{Ar}] 3d^{10} 4s^2 4p^2$ (3) $[\text{Kr}] 3d^{10} 4s^2 4p^2$ (4) $[\text{Kr}] 4d^{10} 5s^2 5p^2$
43. Which one of the following statements for D.I. Mendeleeff, is incorrect ? [JEE-Main-Online 2021]
(1) Element with atomic number 101 is named after him.
(2) He authored the textbook - Principles of Chemistry.
(3) He invented accurate barometer.
(4) At the time, he proposed Periodic Table of elements structure of atom was known.

44. The ionic radii of K^+ , Na^+ , Al^{3+} and Mg^{2+} are in the order : [JEE-Main-Online 2021]
 (1) $Na^+ < K^+ < Mg^{2+} < Al^{3+}$ (2) $Al^{3+} < Mg^{2+} < Na^+ < K^+$
 (3) $Al^{3+} < Mg^{2+} < K^+ < Na^+$ (4) $K^+ < Al^{3+} < Mg^{2+} < Na^+$
45. The ionic radii of F^- and O^{2-} respectively are 1.33 Å and 1.4 Å, while the covalent radius of N is 0.74 Å. The correct statement for the ionic radius of N^{3-} from the following is : [JEE-Main-Online 2021]
 (1) It is bigger than F^- and N, but smaller than of O^{2-}
 (2) It is bigger than O^{2-} and F^-
 (3) It is smaller than O^{2-} and F^- , but bigger than of N
 (4) It is smaller than F^- and N
46. The CORRECT order of first ionisation enthalpy is : [JEE-Main-Online 2021]
 (1) $Mg < Al < S < P$ (2) $Mg < Al < P < S$
 (2) $Al < Mg < S < P$ (4) $Mg < S < Al < P$
47. The correct order of ionic radii for the ions, P^{3-} , S^{2-} , Ca^{2+} , K^+ , Cl^- is : [JEE-Main-Online 2021]
 (1) $K^+ > Ca^{2+} > P^{3-} > S^{2-} > Cl^-$ (2) $P^{3-} > S^{2-} > Cl^- > Ca^{2+} > K^+$
 (3) $P^{3-} > S^{2-} > Cl^- > K^+ > Ca^{2+}$ (4) $Cl^- > S^{2-} > P^{3-} > Ca^{2+} > K^+$
48. The Eu^{2+} ion is a strong reducing agent in spite of its ground state electronic configuration (outermost) : [Atomic number of Eu = 63] [JEE-Main-Online 2021]
 (1) $4f^6$ (2) $4f^6 6s^2$ (3) $4f^7$ (4) $4f^7 6s^2$
49. Given below are two statements : one is labelled as **Assertion (A)** and the other is labelled as Reason (R). [JEE-Main-Online 2021]
Assertion (A) : Metallic character decreases and non-metallic character increases on moving from left to right in a period.
Reason (R) : It is due to increase in ionisation enthalpy and decrease in electron gain enthalpy, when one moves from left to right in a period.
 In the light of the above statements, choose the most appropriate answer from the options given below :
 (1) (A) false but (R) is true
 (2) Both (A) and (R) are correct but (R) is not the correct explanation of (A)
 (3) Both (A) and (R) are correct and (R) is the correct explanation of (A)
 (4) (A) is true but (R) is false
50. Which one of the following lanthanides exhibits +2 oxidation state with diamagnetic nature? (Given Z for Nd = 60, Yb = 70, La = 57, Ce = 58) [JEE-Main-Online 2021]
 (1) La (2) Ce (3) Nd (4) Yb
51. Identify the element for which electronic configuration in +3 oxidation state is $[Ar]3d^5$: [JEE-Main-Online 2021]
 (1) Mn (2) Fe (3) Co (4) Ru
52. The Azimuthal quantum number for the valence electrons of Ga^+ ion is _____. (Atomic number of Ga = 31) [JEE-Main-Online 2021]
53. Number of electrons that Vanadium (Z = 23) has in p-orbitals is equal to _____. [JEE-Main-Online 2021]
54. The number of 5f electrons in the ground state electronic configuration of Np (Z = 93) is..... (Integer answer) [JEE-Main-Online 2021]

55. The value of magnetic quantum number of the outermost electron of Zn^+ ion is _____. (Integer answer)
[JEE-Main-Online 2021]
56. Number of electrons present in 4f orbital of Ho^{3+} ion is _____. (Given Atomic No. of Ho = 67)
[JEE-Main-Online 2021]
57. The number of 4f electrons in the ground state electronic configuration of Gd^{2+} is _____.
[Atomic number of Gd = 64]
[JEE-Main-Online 2021]
58. The correct order of increasing ionic radii is :
(1A) $\text{Mg}^{2+} < \text{Na}^+ < \text{F}^- < \text{O}^{2-} < \text{N}^{3-}$ (2) $\text{N}^{3-} < \text{O}^{2-} < \text{F}^- < \text{Na}^+ < \text{Mg}^{2+}$
(3) $\text{F}^- < \text{Na}^+ < \text{O}^{2-} < \text{Mg}^{2+} < \text{N}^{3-}$ (4) $\text{Na}^+ < \text{F}^- < \text{Mg}^{2+} < \text{O}^{2-} < \text{N}^{3-}$
[JEE-Main-Online 2022]
59. Consider the following statements :
(A) The principal quantum number 'n' is a positive integer with values of 'n' = 1, 2, 3,....
(B) The azimuthal quantum number ℓ for a given 'n' (principal quantum number) can have values as $\ell = 0, 1, 2, \dots, n$
(C) Magnetic orbital quantum number ' m_ℓ ' for a particular ℓ (azimuthal quantum number) has $(2\ell + 1)$ values.
(D) $+1/2$ are the two possible orientations of electron spin.
(E) For $\ell = 5$, there will be a total of 9 orbital
Which of the above statements are correct?
(1) (A), (B) and (C) (2) (A), (C), (D) and (E)
(3) (A), (C) and (D) (4) (A), (B), (C) and (D)
[JEE-Main-Online 2022]
60. Element "E" belongs to the period 4 and group 16 of the periodic table. The valence shell electron configuration of the element, which is just above "E" in the group is :
(1) $3s^2, 3p^4$ (2) $3d^{10}, 4s^2, 4p^4$ (3) $4d^{10}, 5s^2, 5p^4$ (4) $2s^2, 2p^4$
[JEE-Main-Online 2022]
61. Which of the following elements is considered as a metalloid?
(1) Sc (2) Pb (3) Bi (4) Te
[JEE-Main-Online 2022]
62. The electronic configuration of Pt (atomic number 78) is :
(1) $[\text{Xe}] 4f^{14} 5d^9 6s^1$ (2) $[\text{Kr}] 4f^{14} 5d^{10}$ (3) $[\text{Xe}] 4f^{14} 5d^{10}$ (4) $[\text{Xe}] 4f^{14} 5d^8 6s^2$
[JEE-Main-Online 2022]
63. Which of the following statements are correct?
(a) The electronic configuration of Cr is $[\text{Ar}] 3d^5 4s^1$.
(b) The magnetic quantum number may have a negative value
(c) In the ground state of an atom, the orbitals are filled in order of their increasing energies
(d) The total number of nodes are given by $n - 2$
Choose the most appropriate answer from the options given below :
(1) (a), (c) and (d) only (2) (a) and (b) only (3) (a) and (c) only (4) (a), (b) and (c) only
[JEE-Main-Online 2022]
64. The pair, in which ions are isoelectronic with Al^{3+} is :
(1) Br and Be^{2+} (2) Cl^- and Li^+ (3) S^{2-} and K^+ (4) O^{2-} and Mg^{2+}
[JEE-Main-Online 2022]
65. The correct order of electron gain enthalpies of Cl, F, Te and Po is :
(1) $\text{F} < \text{Cl} < \text{Te} < \text{Po}$ (2) $\text{Po} < \text{Te} < \text{F} < \text{Cl}$
(3) $\text{Te} < \text{Po} < \text{Cl} < \text{F}$ (4) $\text{Cl} < \text{F} < \text{Te} < \text{Po}$
[JEE-Main-Online 2022]

66. Consider the following pairs of electrons

[JEE-Main-Online 2022]

- (A) (a) $n = 3, l = 1, m_l = 1, m_s = +\frac{1}{2}$; (b) $n = 3, l = 2, m_l = 1, m_s = +\frac{1}{2}$
- (B) (a) $n = 3, l = 2, m_l = -2, m_s = \frac{1}{2}$; (b) $n = 3, l = 2, m_l = -1, m_s = -\frac{1}{2}$
- (C) (a) $n = 4, l = 2, m_l = 2, m_s = +\frac{1}{2}$; (b) $n = 3, l = 2, m_l = 2, m_s = +\frac{1}{2}$

The pairs of electrons present in degenerate orbitals is /are :

- (1) Only (A) (2) Only (B) (3) Only (C) (4) (B) and (C)

67. Given below are two statements. One is labelled as Assertion A and the other is labelled as Reason R.

Assertion A: The first ionization enthalpy for oxygen is lower than that of nitrogen.

Reason R : The four electrons in 2p orbitals of oxygen experience more electron-electron repulsion.

In the light of the above statements, choose the correct answer from the options given below.

- (1) Both A and R are correct and R_j is the correct explanation of A
(2) Both A and R are correct but R is NOT the correct explanation of A
(3) A is correct but R is not correct
(4) A is not correct but R is correct

68. Given below are two statements : one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : The ionic radii of O^{2-} and Mg^{2+} are same.

Reason (R) : Both O^{2-} and Mg^{2+} are isoelectronic species.

[JEE-Main-Online 2022]

In the light of the above statements, choose the **correct** answer from the options given below.

- (1) Both **(A)** and **(R)** are correct and **(R)** is the correct explanation of **(A)**
- (2) Both **(A)** and **(R)** are correct but **(R)** is NOT the correct explanation of **(A)**
- (3) **(A)** is correct but **(R)** is not correct
- (4) **(A)** is not correct but **(R)** is correct

69. Consider the following set of quantum numbers.

	n	l	m_l
A.	3	3	-3
B.	3	2	-2
C.	2	1	+1
D.	2	2	+2

The number of correct sets of quantum numbers is _____.

[JEE-Main-Online 2022]

70. The difference between electron gain enthalpies will be maximum between :

- (1) Ne and F (2) Ar and F [JEE-Main-Online 2023]
(3) Ne and Cl (4) Ar and Cl

71. The correct order of electronegativity for given elements is :

[JEE-Main-Online 2023]

- (1) P > Br > C > At
 (2) Br > P > At > C
 (3) Br > C > At > P
 (4) C > P > At > Br

72. Which of the following statements are not correct? [JEE-Main-Online 2023]
 A. The electron gain enthalpy of F is more negative than that of Cl.
 B. Ionization enthalpy decreases in a group of periodic table.
 C. The electronegativity of an atom depends upon the atoms bonded to it.
 D. Al_2O_3 and NO are examples of amphoteric oxides.
 Choose the most appropriate answer from the options given below :
 (1) B and D Only (2) A, B, C and D
 (3) A, B and D Only (4) A, C and D Only
73. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R
Assertion A : The energy required to form Mg^{2+} from Mg is much higher than that required to produce Mg^+
Reason R : Mg^{2+} is small ion and carry more charge than Mg^+
 In the light of the above statements, choose the correct answer from the options given below.
 (1) A is true but R is false [JEE-Main-Online 2023]
 (2) A is false but R is true
 (3) Both A and R are true and R is the correct explanation of A
 (4) Both A and R are true but R is NOT the correct explanation of A
74. For elements B, C, N, Li, Be, O and F, the correct order of first ionization enthalpy is
 (1) $\text{Li} < \text{B} < \text{Be} < \text{C} < \text{O} < \text{N} < \text{F}$ (2) $\text{Li} < \text{Be} < \text{B} < \text{C} < \text{O} < \text{N} < \text{F}$ [JEE-Main-Online 2023]
 (3) $\text{B} > \text{Li} > \text{Be} > \text{C} > \text{N} > \text{O} > \text{F}$ (4) $\text{Li} < \text{Be} < \text{B} < \text{C} < \text{N} < \text{O} < \text{F}$
75. The number of atomic orbitals from the following having 5 radial nodes is _____.
 7s, 7p, 6s, 8p, 8d [JEE-Main-Online 2023]
76. For a metal ion, the calculated magnetic moment is 4.90 BM. This metal ion has _____ number of unpaired electrons.
 [JEE-Main-Online 2023]
77. For electron gain enthalpies of the elements denoted as $\Delta_{\text{eg}} H$, the correct option is :
 (1) $\Delta_{\text{eg}} H(\text{Cl}) < \Delta_{\text{eg}} H(\text{F})$ [JEE-Main-Online 2023]
 (2) $\Delta_{\text{eg}} H(\text{Se}) < \Delta_{\text{eg}} H(\text{S})$
 (3) $\Delta_{\text{eg}} H(\text{Te}) < \Delta_{\text{eg}} H(\text{Po})$
 (4) $\Delta_{\text{eg}} H(\text{I}) < \Delta_{\text{eg}} H(\text{At})$
78. The correct increasing order of the ionic radii is
 (1) $\text{Cl}^- < \text{Ca}^{2+} < \text{K} < \text{S}^{2-}$ (2) $\text{K}^+ < \text{S}^{2-} < \text{Ca}^{2+} < \text{Cl}^-$ [JEE-Main-Online 2023]
 (3) $\text{Ca}^{2+} < \text{K}^+ < \text{Cl}^- < \text{S}^{2-}$ (4) $\text{S}^{2-} < \text{Cl}^- < \text{Ca}^{2+} < \text{K}^+$
79. Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R).
Assertion (A) : The first ionization enthalpy of 3d series elements is more than that of group 2 metals
Reason (R) : In 3d series of elements successive filling of d-orbitals takes place.
 In the light of the above statements, choose the correct answer from the options given below:
 (1) Both (A) and (R) are true but (R) is not the correct explanation of (A)
 (2) Both (A) and (R) are true and (R) is the correct explanation of (A) [JEE-Main-Online 2023]
 (3) (A) is false but (R) is true
 (4) (A) is true but (R) is false

80. Which of the following elements have half-filled f-orbitals in their ground state?
(Given : atomic number Sm = 62; Eu = 63; Tb = 65; Gd = 64, Pm = 61) [JEE-Main-Online 2023]
A. Sm B. Eu C. Tb D. Gd E. Pm
Choose the correct answer from the options given below :
(1) A and B only (2) B and D only (3) A and E only (4) C and D only
81. $\text{Nd}^{2+} =$ _____ [JEE-Main-Online 2023]
(1) $4f^4$ (2) $4f^3$ (3) $4f^4 6s^2$ (4) $4f^2 6s^2$
82. Maximum number of electrons that can be accommodated in shell with $n = 4$ are :
(1) 16 (2) 32 (3) 72 (4) 50 [JEE-Main-Online 2023]
83. Match List-I with List-II
- | List-I
(Atomic number) | List-II
(Block of periodic table) |
|---------------------------|--------------------------------------|
| (A) 37 | (I) p-block |
| (B) 78 | (II) d-block |
| (C) 52 | (III) f-block |
| (D) 65 | (IV) s-block |
- Choose the correct answer from the options given below : [JEE-Main-Online 2023]
(1) A-IV B-III C-II D-I
(2) A-II B-IV C-I D-III
(3) A-IV B-II C-I D-III
(4) A-I B-III C-IV D-II
84. Which of the following represents the correct order of metallic character of the given elements?
(1) $\text{Be} < \text{Si} < \text{K} < \text{Mg}$ (2) $\text{Be} < \text{Si} < \text{Mg} < \text{K}$ [JEE-Main-Online 2023]
(3) $\text{Si} < \text{Be} < \text{Mg} < \text{K}$ (4) $\text{K} < \text{Mg} < \text{Be} < \text{Si}$
85. Inert gases have positive electron gain enthalpy. Its correct order is
(1) $\text{He} < \text{Xe} < \text{Kr} < \text{Ne}$ (2) $\text{He} < \text{Ne} < \text{Kr} < \text{Xe}$ [JEE-Main-Online 2023]
(3) $\text{He} < \text{Kr} < \text{Xe} < \text{Ne}$ (4) $\text{Xe} < \text{Kr} < \text{Ne} < \text{He}$
86. Which one of the following sets of ions represents a collection of isoelectronic species?
(Given : Atomic Number : F : 9, Cl : 17, Na = 11, Mg = 12, Al = 13, K = 19, Ca = 20, Sc = 21)
(1) $\text{N}^{3-}, \text{O}^{2-}, \text{F}^-, \text{S}^{2-}$ (2) $\text{Ba}^{2+}, \text{Sr}^{2+}, \text{K}^+, \text{Ca}^{2+}$ [JEE-Main-Online 2023]
(3) $\text{K}^+, \text{Cl}^-, \text{Ca}^{2+}, \text{Sc}^{3+}$ (4) $\text{Li}^+, \text{Na}^+, \text{Mg}^{2+}, \text{Ca}^{2+}$
87. Arrange the following orbitals in decreasing order of energy. [JEE-Main-Online 2023]
A. $n = 3, l = 0, m = 0$
B. $n = 4, l = 0, m = 0$
C. $n = 3, l = 1, m = 0$
D. $n = 3, l = 2, m = 1$
The correct option for the order is :
(1) $\text{D} > \text{B} > \text{A} > \text{C}$ (2) $\text{B} > \text{D} > \text{C} > \text{A}$ (3) $\text{D} > \text{B} > \text{C} > \text{A}$ (4) $\text{A} > \text{C} > \text{B} > \text{D}$

88. The magnetic moment of a transition metal compound has been calculated to be 3.87 B.M. The metal ion is: [JEE-Main-Online 2023]
 (1) V^{2+} (2) Ti^{2+} (3) Cr^{2+} (4) Mn^{2+}
89. The number of s-electrons present in an ion with 55 protons in its unipositive state is [JEE-Main-Online 2023]
 (1) 10 (2) 9 (3) 12 (4) 8
90. How many of the following metal ions have similar value of spin only magnetic moment in gaseous state? [JEE-Main-Online 2023]

 (Given : Atomic number : V, 23; Cr, 24; Fe, 26; Ni, 28)
 V^{3+} , Cr^{3+} , Fe^{2+} , Ni^{3+}
91. The number of alkali metal(s), from Li, K, Cs, Rb having ionization enthalpy greater than 400 kJ mol^{-1} and forming stable super oxide is _____. [JEE-Main-Online 2023]
92. The electronic configuration for Neodymium is [JEE-Main-Online 2024]
 [Atomic Number for Neodymium 60]
 (1) $[Xe] 4f^4 6s^2$ (2) $[Xe] 4f^6 6s^2$ (3) $[Xe] 5f^7 7s^2$ (4) $[Xe] 4f^1 5d^1 6s^2$
93. In case of isoelectronic species the size of F^- , Ne and Na^+ is affected by [JEE-Main-Online 2024]
 (1) Nuclear charge (z)
 (2) Principal quantum number (n)
 (3) None of the factors because their size is the same
 (4) Electron-electron interaction in the outer orbitals
94. The transition metal having highest 3rd ionisation enthalpy is [JEE-Main-Online 2024]
 (1) Cr (2) Mn (3) V (4) Fe
95. The four quantum numbers for the outer most orbital of potassium (atomic no. 19) are [JEE-Main-Online 2024]
 (1) $n = 4, \ell = 0, m = 0, s = +\frac{1}{2}$ (2) $n = 2, \ell = 0, m = 0, s = +\frac{1}{2}$
 (3) $n = 4, \ell = 2, m = -1, s = +\frac{1}{2}$ (4) $n = 3, \ell = 0, m = 1, s = +\frac{1}{2}$
96. The correct sequence of electron affinity of the elements listed below is : [JEE-Main-Online 2024]
 (A) Ar (B) Br (C) F (D) S
 Choose the **most appropriate** from the options given below :
 (1) $C > B > D > A$ (2) $A > D > C > B$ (3) $D > C > B > A$ (4) $A > D > B > C$
97. The element having the highest first ionization enthalpy is [JEE-Main-Online 2024]
 (1) Al (2) Si (3) C (4) N
98. Which of the following electronic configuration would be associated with the highest magnetic moment? [JEE-Main-Online 2024]
 (1) $[Ar] 3d^3$ (2) $[Ar] 3d^6$ (3) $[Ar] 3d^7$ (4) $[Ar] 3d^8$

99. Consider the following elements :

Group	A' B' →	Period
	C' D'	

Which of the following is/are true about A', B', C' and D'?

[JEE-Main-Online 2024]

- (a) Order of atomic radii : $B' < A' < D' < C'$
 (b) Order of metallic character : $B' < A' < D' < C'$
 (c) Size of the element : $D' < C' < B' < A'$
 (d) Order of ionic radii : $B'^+ < A'^+ < D'^+ < C'^+$

Choose the correct answer from the options given below :

- (1) b, c and d only (2) a and b only (3) a, b and d only (4) a only

100. Choose the correct option having all the elements with d^{10} electronic configuration from the following :

- (1) ^{46}Pd , ^{28}Ni , ^{26}Fe , ^{24}Cr (2) ^{27}Co , ^{28}Ni , ^{26}Fe , ^{24}Cr [JEE-Main-Online 2024]
 (3) ^{29}Cu , ^{30}Zn , ^{48}Cd , ^{47}Ag (4) ^{28}Ni , ^{24}Cr , ^{26}Fe , ^{29}Cu

101. The correct set of four quantum numbers for the valence electron of rubidium atom ($Z = 37$) is

[JEE-Main-Online 2024]

- (1) 5, 1, 0, $+\frac{1}{2}$ (2) 5, 0, 1, $+\frac{1}{2}$ (3) 5, 0, 0, $+\frac{1}{2}$ (4) 5, 1, 1, $+\frac{1}{2}$

102. Given below are two statements :

Statement-I : Fluorine has most negative electron gain enthalpy in its group.

Statement-II : Oxygen has least negative electron gain enthalpy in its group.

In the light of the above statements, choose the most appropriate answer from the options given below :

- (1) Both Statement-I and Statement-II are true
 (2) Statement-I is false but Statement-II is true
 (3) Statement-I is true but Statement-II is false
 (4) Both Statement-I and Statement-II is false

[JEE-Main-Online 2024]

103. Given below are two statements :

Statement (I) : The orbitals having same energy are called as degenerated orbitals

Statement (II) : In hydrogen atom, 3p and 3d orbitals are not degenerate orbitals.

In the light of the above statements, choose the most appropriate answer from the options given below :

- (1) Both **Statement I** and **Statement II** are false
 (2) **Statement I** is true but **Statement II** is false
 (3) **Statement I** is false but **Statement II** is true
 (4) Both **Statement I** and **Statement II** are true

[JEE-Main-Online 2024]

104. Given below are two statements : One is labelled as Assertion A and the other is labelled as Reason R :

Assertion A : The first ionisation enthalpy decreases across a period.

Reason R : The increasing nuclear charge outweighs the shielding across the period.

In the light of the above statements, choose the most appropriate from the options given below :

- (1) Both **A** and **R** are true and **R** is the correct explanation of **A**
 (2) **A** is true but **R** is false
 (3) Both **A** and **R** are true but **R** is NOT the correct explanation of **A**
 (4) **A** is false but **R** is true

[JEE-Main-Online 2024]

105. Given below are two statements :

Statement-I : The 4f and 5f-series of elements are placed separately in the Periodic table to preserve the principle of classification.

Statement-II : s-block elements can be found in pure form in nature. **[JEE-Main-Online 2024]**

In the light of the above statements, choose the most appropriate answer from the options given below :

- (1) Both Statement-I and Statement-II are false
(2) Both Statement-I and Statement-II are true
(3) Statement-I is false but Statement-II is true
(4) Statement-I is true but Statement-II is false

106. Match List - I with List - II

[JEE-Main-Online 2024]

List - I		List - II	
	Species		Electronic distribution
(A)	Cr^{+2}	(I)	$3d^8$
(B)	Mn^{+}	(II)	$3d^3 4s^1$
(C)	Ni^{+2}	(III)	$3d^4$
(D)	V^{+}	(IV)	$3d^5 4s^1$

Choose the correct answer from the options given below :

- (1) (A) – (I), (B) – (II), (C) – (III), (D) – (IV)
 (2) (A) – (II), (B) – (I), (C) – (IV), (D) – (III)
 (3) (A) – (III), (B) – (IV), (C) – (I), (D) – (II)
 (4) (A) – (IV), (B) – (III), (C) – (I), (D) – (II)

107. Arrange the following elements in the increasing order of number of unpaired electrons in it.

- (i) Sc (ii) Cr (iii) V (iv) Ti
- (v) Mn

Choose the correct answer from the options given below :

[JEE-Main-Online 2024]

- (1) (iii) < (v) < (ii) < (i) < (iv) (2) (ii) < (iii) < (iv) < (v) < (i)
(3) (i) < (iv) < (iii) < (v) < (ii) (4) (i) < (iv) < (iii) < (ii) < (v)

108. The electron gain enthalpy value are negative for

- (a) $\text{Be} \rightarrow \text{Be}^-$ (b) $\text{N} \rightarrow \text{N}^-$
(c) $\text{O} \rightarrow \text{O}^{2-}$ (d) $\text{Na} \rightarrow \text{Na}^-$
(e) $\text{Al} \rightarrow \text{Al}^-$

Choose the most appropriate answer from the options given below :

[JEE-Main-Online 2024]

- (1) a and d only (2) a, b and c only (3) d and e only (4) a, b, d and e only

109. The correct order of the first ionization enthalpy is :

[JEE-Main-Online 2024]

- (1) $B > Al > Ga$ (2) $Ga > Al > B$ (3) $Tl > Ga > Al$ (4) $Al > Ga > Tl$

110. The correct order of first ionization enthalpy values of the following elements is :

- (A) O (B) N (C) Be (D) F (E) B

Choose the correct answer from the options given below :

[JEE-Main-Online 2024]

- (1) $E < C < A < B < D$
 (2) $B < D < C < E < A$
 (3) $C < E < A < B < D$
 (4) $A < B < D < C < E$

111. Given below are two statements :

Statement-I : The metallic radius of Na is 1.86 Å and the ionic radius of Na⁺ is lesser than 1.86 Å.

Statement-II : Ions are always smaller in size than the corresponding elements.

In the light of the above statements, choose the **correct** answer from the options given below :

[JEE-Main-Online 2024]

- (1) Statement I is incorrect but Statement II is true
- (2) Statement I is correct but Statement II is false
- (3) Both Statement I and Statement II are false
- (4) Both Statement I and Statement II are true

112. Given below are two statements :

Statement-I : The correct order of first ionization enthalpy values of Li, Na, F and Cl is Na < Li < Cl < F.

Statement-II : The correct order of negative electron gain enthalpy values of Li, Na, F and Cl is Na < Li < F < Cl

In the light of the above statements, choose the correct answer from the options given below :

[JEE-Main-Online 2024]

- (1) Both Statement I and Statement II are false
- (2) Both Statement I and Statement II are true
- (3) Statement I is true but Statement II is false
- (4) Statement I is false but Statement II is true

113. Match List I with List II

List I

A. N

B. S

C. Br

D. Kr

List II

I. [Ar] 3d¹⁰ 4s² 4p⁵

II. [Ne] 3s² 3p⁴

III. [He] 2s² 2p³

IV. [Ar] 3d¹⁰ 4s² 4p⁶

Choose the correct answer from the options given below :

[JEE-Main-Online 2024]

(1) A-II, B-I, C-IV, D-III

(2) A-III, B-II, C-I, D-IV

(3) A-IV, B-III, C-II, D-II

(4) A-I, B-IV, C-III, D-II

114. Compare the energies of following sets of quantum numbers for multielectron system.

(A) n = 4, l = 1

(B) n = 4, l = 2

(C) n = 3, l = 1

(D) n = 3, l = 2

(E) n = 4, l = 0

[JEE-Main-Online 2024]

(1) (E) < (C) < (D) < (A) < (B)

(2) (E) > (C) > (A) > (D) > (B)

(3) (C) < (E) < (D) < (A) < (B)

(4) (B) > (A) > (C) > (E) > (D)

115. The number of electrons present in all the complete filled subshells having n = 4 and s = + $\frac{1}{2}$ is _____.

(Where n = principal quantum number and s = spin quantum number)

[JEE-Main-Online 2024]

116. Total number of ions from the following with noble gas configuration is _____. [JEE-Main-Online 2024]
Sr²⁺ (z = 38), Cs⁺ (z = 55), La²⁺ (z = 57), Pb²⁺ (z = 82), Yb²⁺ (z = 70) and Fe²⁺ (z = 26)

117. If IUPAC name of an element is "Unununnium" then the element belongs to nth group of Periodic table.
The value of n is _____.

[JEE-Main-Online 2024]

118. The maximum number of orbitals which can be identified with n = 4 and m_l = 0 is _____.

[JEE-Main-Online 2024]

119. Which of the following electronegativity order is **incorrect**? [JEE-Main-Online 2025]
 (1) $\text{Al} < \text{Mg} < \text{B} < \text{N}$ (2) $\text{Al} < \text{Si} < \text{C} < \text{N}$ (3) $\text{Mg} < \text{Be} < \text{B} < \text{N}$ (4) $\text{S} < \text{Cl} < \text{O} < \text{F}$
120. Lanthanoid ions with $4f^7$ configuration are :
 (A) Eu^{2+} (B) Gd^{3+} (C) Eu^{3+} (D) Tb^{3+} (E) Sm^{2+}
 Choose the correct answer from the options given below : [JEE-Main-Online 2025]
 (1) (A) and (B) only (2) (A) and (D) only
 (3) (B) and (E) only (4) (B) and (C) only
121. Given below are two statements :
Statement (I) : According to the Law of Octaves, the elements were arranged in the increasing order of their atomic number.
Statement (II) : Meyer observed a periodically repeated pattern upon plotting physical properties of certain elements against their respective atomic numbers.
 In the light of the above statements, choose the **correct** answer from the options given below : [JEE-Main-Online 2025]
 (1) Statement I is false but Statement II is true
 (2) Both Statement I and Statement II are true
 (3) Statement I is true but Statement II is false
 (4) Both Statement I and Statement II are false
122. Given below are two statements :
Statement (I) : The radii of isoelectronic species increases in the order.
 $\text{Mg}^{2+} < \text{Na}^+ < \text{F}^- < \text{O}^{2-}$
Statement (II) : The magnitude of electron gain enthalpy of halogen decreases in the order.
 $\text{Cl} > \text{F} > \text{Br} > \text{I}$
 In the light of the above statements, choose the **most appropriate answer** from the options given below : [JEE-Main-Online 2025]
 (1) Statement I is incorrect but Statement II is correct
 (2) Both Statement I and Statement II are incorrect
 (3) Statement I is correct but Statement II is incorrect
 (4) Both Statement I and Statement II are correct
123. Which of the following is/are not correct with respect to energy of atomic orbitals of hydrogen atom?
 (A) $1s < 2p < 3d < 4s$ (B) $1s < 2s = 2p < 3s = 3p$
 (C) $1s < 2s < 2p < 3s < 3p$ (D) $1s < 2s < 4s < 3d$
 Choose the correct answer from the options given below : [JEE-Main-Online 2025]
 (1) (B) and (D) only (2) (A) and (C) only (3) (C) and (D) only (4) (A) and (B) only
124. The element that does not belong to the same period of the remaining elements (modern periodic table) is [JEE-Main-Online 2025]
 (1) Palladium (2) Iridium (3) Osmium (4) Platinum
125. The successive 5 ionisation energies of an element are 800, 2427, 3658, 25024 and 32824 kJ/mol, respectively. By using the above values predict the group in which the above element is present : [JEE-Main-Online 2025]
 (1) Group 2 (2) Group 13 (3) Group 4 (4) Group 14

126. In a multielectron atom, which of the following orbitals described by three quantum numbers with have same energy in absence of electric and magnetic fields?
- A. $n = 1, l = 0, m_l = 0$
 B. $n = 2, l = 0, m_l = 0$
 C. $n = 2, l = 1, m_l = 1$
 D. $n = 3, l = 2, m_l = 1$
 E. $n = 3, l = 2, m_l = 0$
- Choose the correct answer from the options given below : [JEE-Main-Online 2025]
- (1) A and B only (2) B and C only (3) C and D only (4) D and E only
127. The incorrect decreasing order of atomic radii is : [JEE-Main-Online 2025]
- (1) $\text{Mg} > \text{Al} > \text{C} > \text{O}$ (2) $\text{Al} > \text{B} > \text{N} > \text{F}$ (3) $\text{Be} > \text{Mg} > \text{Al} > \text{Si}$ (4) $\text{Si} > \text{P} > \text{Cl} > \text{F}$
128. For hydrogen atom, the orbital/s with lowest energy is/are :
- (A) $4s$ (B) $3p_x$ (C) $3d_{x^2-y^2}$ (D) $3d_{z^2}$ (E) $4p_z$
- Choose the correct answer from the options given below : [JEE-Main-Online 2025]
- (1) (A) and (E) only (2) (B) only (3) (A) only (4) (B), (C) and (D) only
129. Which of the following Statements are NOT true about the periodic table?
- (A) The properties of elements are function of atomic weights.
 (B) The properties of elements are function of atomic numbers.
 (C) Elements having similar outer electronic configuration are arranged in same period.
 (D) An element's location reflects the quantum numbers of the last filled orbital.
 (E) The number of elements in a period is same as the number of atomic orbitals available in energy level that is being filled.
- Choose the correct answer from the options given below : [JEE-Main-Online 2025]
- (1) A, C and E Only (2) D and E Only
 (3) A and E Only (4) B, C and E Only
130. Given below are two statements :
- Statement (I)** : The first ionization energy of Pb is greater than that of Sn.
Statement (II) : The first ionization energy of Ge is greater than that of Si.
- In the light of the above statements, choose the correct answer from the options given below : [JEE-Main-Online 2025]
- (1) Statement I is true but Statement II is false (2) Both Statement I and Statement II are false
 (3) Statement I is false but Statement II is true (4) Both Statement I and Statement II are true
131. Given below are two statements :
- Statement (I)** : An element in the extreme left of the periodic table forms acidic oxides.
Statement (II) : Acid is formed during the reaction between water and oxide of a reactive element present in the extreme right of the periodic table.
- In the light of the above statements, choose the correct answer from the options given below : [JEE-Main-Online 2025]
- (1) Statement-I is false but Statement-II is true.
 (2) Both Statement-I and Statement-II are false.
 (3) Statement-I is true but Statement-II is false.
 (4) Both Statement-I and Statement-II are true.

132. Match List-I with List-II.

List-I (Transition metal ion)	List-II (Spin only magnetic moment (B.M.))
(A) Ti^{3+}	(I) 3.87
(B) V^{2+}	(II) 0.00
(C) Ni^{2+}	(III) 1.73
(D) Sc^{3+}	(IV) 2.84

Choose the **correct** answer from the options given below :

[JEE-Main-Online 2025]

- | | |
|--|--|
| (1) (A)-(III), (B)-(I), (C)-(II), (D)-(IV) | (2) (A)-(III), (B)-(I), (C)-(IV), (D)-(II) |
| (3) (A)-(IV), (B)-(II), (C)-(III), (D)-(I) | (4) (A)-(II), (B)-(IV), (C)-(I), (D)-(III) |

133. Given below are two statements :

Statement (I) : For a given shell, the total number of allowed orbitals is given by n^2 .

Statement (II) : For any subshell, the spatial orientation of the orbitals is given by $-\ell$ to $+\ell$ values including zero.

In the light of the above statements, choose the **correct** answer from the options given below :

[JEE-Main-Online 2025]

- (1) Statement I is true but Statement II is false
- (2) Statement I is false but Statement II is true
- (3) Both Statement I and Statement II are true
- (4) Both Statement I and Statement II are false

134. Given below are two statements :

Statement (I) : A spectral line will be observed for a $2p_x \rightarrow 2p_y$ transition.

Statement (II) : $2p_x$ and $2p_y$ are degenerate orbitals.

In the light of the above statements, choose the correct answer from the options given below :

[JEE-Main-Online 2025]

- | | |
|---|---|
| (1) Both Statement-I and Statement-II are true. | (2) Both Statement-I and Statement-II are false. |
| (3) Statement-I is true but Statement-II is false | (4) Statement-I is false but Statement-II is true |

135. Given below are two statements about X-ray spectra of elements :

Statement (I) : A plot of $\sqrt{\nu}$ (ν = frequency of X-rays emitted) vs atomic mass is a straight line.

Statement (II) : A plot of ν (ν = frequency of X-rays emitted) vs atomic number is a straight line.

In the light of the above statements choose the **correct** answer from the options given below :

[JEE-Main-Online 2025]

- | | |
|---|---|
| (1) Statement I is true but Statement II is false | (2) Both Statement I and Statement II are true |
| (3) Both Statement I and Statement II are false | (4) Statement I is false but Statement II is true |

136. Match List-I with List-II

List-I	List-II
(A) $Al^{3+} < Mg^{2+} < Na^+ < F^-$	(I) Ionisation Enthalpy
(B) $B < C < O < N$	(II) Metallic character
(C) $B < Al < Mg < K$	(III) Electronegativity
(D) $Si < P < S < Cl$	(IV) Ionic radii

Choose the correct answer from the options given below :

[JEE-Main-Online 2025]

- | | |
|----------------------------|----------------------------|
| (1) A-IV, B-I, C-III, D-II | (2) A-II, B-III, C-IV, D-I |
| (3) A-IV, B-I, C-II, D-III | (4) A-III, B-IV, C-II, D-I |

EXERCISE-IV

(Question asked in previous IIT-JEE)

1. Moving from right to left in a periodic table, the atomic size is : [JEE 1995]
(1) increased (2) decreased
(3) remains constant (4) none of these
2. The increasing order of electronegativity in the following elements : [JEE 1995]
(1) C, N, Si, P (2) N, Si, C, P (3) Si, P, C, N (4) P, Si, N, C
3. One element has atomic weight 39. Its electronic configuration is $1s^2, 2s^2 2p^6, 3s^2 3p^6 4s^1$. The true statement for that element is : [JEE 1995]
(1) High value of IE (2) Transition element (3) Isotone with ${}_{18}\text{Ar}^{38}$ (4) None
4. The number of paired electrons in oxygen is : [JEE 1995]
(1) 6 (2) 16 (3) 8 (4) 32
5. The decreasing size of K^+ , Ca^{2+} , Cl^- & S^{2-} follows the order : [JEE 1995]
(1) $\text{K}^+ > \text{Ca}^{2+} > \text{S}^{2-} > \text{Cl}^-$ (2) $\text{K}^+ > \text{Ca}^{2+} > \text{Cl}^- > \text{S}^{2-}$
(3) $\text{Ca}^{2+} > \text{K}^+ > \text{Cl}^- > \text{S}^{2-}$ (4) $\text{S}^{2-} > \text{Cl}^- > \text{K}^+ > \text{Ca}^{2+}$
6. Which of the following oxide is neutral ? [JEE 1996]
(1) CO (2) SnO_2 (3) ZnO (4) SiO_2
7. Which of the following has the maximum number of unpaired electrons [JEE 1996]
(1) Mg^{2+} (2) Ti^{3+} (3) V^{3+} (4) Fe^{2+}
8. The following acids have been arranged in the order of decreasing acid strength. Identify the correct order [JEE 1996]
 ClOH(I) BrOH(II) IOH(III)
(1) $\text{I} > \text{II} > \text{III}$ (2) $\text{II} > \text{I} > \text{III}$ (3) $\text{III} > \text{II} > \text{I}$ (4) $\text{I} > \text{III} > \text{II}$
9. The incorrect statement among the following is : [JEE 1997]
(1) the first ionisation potential of Al is less than the first ionisation potential of Mg
(2) the second ionisation potential of Mg is greater than the second ionisation potential of Na
(3) the first ionisation potential of Na is less than the first ionisation potential of Mg
(4) the third ionisation potential of Mg is greater than the third ionisation potential of Al
10. **Assertion** : F atom has a less negative electron gain enthalpy than Cl atom. [JEE 2000]
Reason : Additional electron is repelled more efficiently by 3p electron in Cl atom than by 2p electron in F atom.
(1) If both assertion and reason are CORRECT, but reason is the CORRECT explanation of the assertion.
(2) If both assertion and reason are CORRECT, but reason is not the CORRECT explanation of the assertion.
(3) If assertion is CORRECT but reason is INCORRECT
(4) If assertion is INCORRECT reason is CORRECT.

11. **Assertion** : $\text{Al}(\text{OH})_3$ is amphoteric in nature. [JEE 2000]
Reason : $\text{Al}-\text{O}$ and $\text{O}-\text{H}$ bonds can be broken with equal ease in $\text{Al}(\text{OH})_3$.
 (1) If both assertion and reason are CORRECT, but reason is the CORRECT explanation of the assertion.
 (2) If both assertion and reason are CORRECT, but reason is not the CORRECT explanation of the assertion.
 (3) If assertion is CORRECT but reason is INCORRECT
 (4) If assertion is INCORRECT reason is CORRECT.
12. The correct order of radii is : [JEE 2000]
 (1) $\text{N} < \text{Be} < \text{B}$ (2) F^- , O^{2-} , N^{3-} (3) $\text{Na} < \text{Li} < \text{K}$ (4) $\text{Fe}^{3+} < \text{Fe}^{2+} < \text{Fe}^{4+}$
13. The correct order of acidic strength is [JEE 2000]
 (1) $\text{Cl}_2\text{O}_7 > \text{SO}_3 > \text{P}_4\text{O}_{10}$ (2) $\text{CO}_2 > \text{N}_2\text{O}_5 > \text{SO}_3$
 (3) $\text{Na}_2\text{O} > \text{MgO} > \text{Al}_2\text{O}_3$ (4) $\text{K}_2\text{O} > \text{CaO} > \text{MgO}$
14. The set representing correct order of IP_1 is [JEE 2001]
 (1) $\text{K} > \text{Na} > \text{Li}$ (2) $\text{Be} > \text{Mg} > \text{Ca}$ (3) $\text{B} > \text{C} > \text{N}$ (4) $\text{Fe} > \text{Si} > \text{C}$
15. Identify the least stable ion amongst the following : [JEE 2002]
 (1) Li^- (2) Be^- (3) B^- (4) C^-
16. Identify the correct order of acidic strengths of CO_2 , CuO , CaO , H_2O [JEE 2002]
 (1) $\text{CaO} < \text{CuO} < \text{H}_2\text{O} < \text{CO}_2$ (2) $\text{H}_2\text{O} < \text{CuO} < \text{CaO} < \text{CO}_2$
 (3) $\text{CaO} < \text{H}_2\text{O} < \text{CuO} < \text{CO}_2$ (4) $\text{H}_2\text{O} < \text{CO}_2 < \text{CaO} < \text{CuO}$
17. The increasing order of atomic radii of the following Group 13 elements is [JEE-Advanced 2016]
 (1) $\text{Al} < \text{Ga} < \text{In} < \text{Tl}$ (2) $\text{Ga} < \text{Al} < \text{In} < \text{Tl}$
 (3) $\text{Al} < \text{In} < \text{Ga} < \text{Tl}$ (4) $\text{Al} < \text{Ga} < \text{Tl} < \text{In}$
18. The 1st, 2nd, and 3rd ionization enthalpies, I_1 , I_2 , I_3 , of four atoms with atomic numbers n , $n+1$, $n+2$ and $n+3$, where $n < 10$, are tabulated below. What is the value of n ? [JEE-Advanced 2020]

Atomic number	Ionization Enthalpy (kJ / mol)		
	I_1	I_2	I_3
n	1681	3374	6050
$n+1$	2081	3952	6122
$n+2$	496	4562	6910
$n+3$	738	1451	7733

ANSWER KEY

EXERCISE-I

1.	3	2.	4	3.	2	4.	2	5.	2	6.	2	7.	3
8.	1	9.	1	10.	4	11.	4	12.	1	13.	2	14.	3
15.	3	16.	4	17.	2	18.	2	19.	2	20.	3	21.	2
22.	1	23.	2	24.	2	25.	2	26.	3	27.	4	28.	3
29.	4	30.	3	31.	3	32.	2	33.	1	34.	1	35.	1
36.	3	37.	3	38.	2	39.	3	40.	3	41.	3	42.	4
43.	1	44.	4	45.	2	46.	3	47.	3	48.	4	49.	3
50.	2	51.	1	52.	1	53.	2	54.	4	55.	1	56.	4
57.	1	58.	1	59.	3	60.	2	61.	5.50	62.	5.50	63.	7
64.	5	65.	6	66.	5	67.	2	68.	2				

EXERCISE-II

1.	3	2.	4	3.	1	4.	1	5.	2	6.	2	7.	4
8.	4	9.	2	10.	2	11.	2	12.	2	13.	2	14.	1
15.	2	16.	4	17.	3	18.	2	19.	2	20.	1	21.	2
22.	3	23.	1	24.	2								

EXERCISE-III

1.	4	2.	3	3.	3	4.	3	5.	2	6.	1	7.	2
8.	2	9.	2	10.	2	11.	1	12.	1	13.	2	14.	3
15.	3	16.	2	17.	2	18.	1	19.	2	20.	3	21.	3
22.	2	23.	101.00	24.	1	25.	3	26.	4	27.	1	28.	2
29.	02.00	30.	2	31.	5576 kJ/mol			32.	1	33.	2	34.	4
35.	3	36.	1	37.	2	38.	1	39.	4	40.	2	41.	49
42.	4	43.	4	44.	2	45.	2	46.	3	47.	3	48.	3
49.	4	50.	4	51.	2	52.	0.00	53.	12.00	54.	4.00	55.	0.00
56.	10.00	57.	7.00	58.	1	59.	3	60.	1	61.	4	62.	1
63.	4	64.	4	65.	2	66.	2	67.	2	68.	4	69.	2.00
70.	C	71.	3	72.	4	73.	3	74.	1	75.	03.00	76.	04.00
77.	2	78.	3	79.	2	80.	2	81.	1	82.	2	83.	3
84.	3	85.	1	86.	3	87.	3	88.	1	89.	1	90.	2.00
91.	2.00	92.	1	93.	1	94.	2	95.	1	96.	1	97.	4
98.	2	99.	3	100.	3	101.	3	102.	2	103.	2	104.	4
105.	4	106.	3	107.	3	108.	3	109.	3	110.	1	111.	2
112.	2	113.	2	114.	3	115.	16.00	116.	2.00	117.	11.00	118.	4.0
119.	1	120.	1	121.	4	122.	4	123.	3	124.	1	125.	2
126.	4	127.	3	128.	4	129.	1	130.	1	131.	1	132.	2
133.	3	134.	4	135.	3	136.	3						

EXERCISE-IV

1.	1	2.	3	3.	3	4.	1	5.	4	6.	1	7.	4
8.	1	9.	2	10.	3	11.	3	12.	2	13.	1	14.	2
15.	2	16.	1	17.	2	18.	9						